

AI-Powered Skill Gap Analysis and Career Recommendation System

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ABSTRACT: *Automation, digital transformation and artificial intelligence are transforming the global job market at a rapid rate, but job seeking tools such as aptitude tests, individual career counselling and keyword-based job search boards are not ready to keep up, making it impossible for students, graduates and job-seekers to get a clear idea of how their current skills match those their employers are looking for. This paper reviews and classifies sixteen papers from IEEE-indexed and peer-reviewed literature that have been published covering the following research areas: resumes parsing, collaborative job recommendation, content-based job recommendation, clustering-based employment matching, generative-AI and retrieval-augmented recommendation, skill-gap policy and organizational analysis, congestion- and fairness-aware recommendation, efficient labour-market text encoding, and rule-based recruitment AI. Building on the gaps identified by this review, this paper proposes an AI-powered Skill Gap Analysis and Career Recommendation System which allows the user to manually enter skills or upload a resume as a pdf file for automatic extraction, represents skill profiles using the TF-IDF (Term Frequency-Inverse Document Frequency) vectorization technique, computes the Cosine Similarity between the user and role descriptions and produces a quantified match percentage, applies an independent Random Forest Classifier to identify the most appropriate career role, creates a ranked list of competencies missing in the user's profile, and persists all the analyses, enabling retracing user progress over time. The system is deployed on a React.js, Node.js/Express.js and MongoDB architecture with a Python/Scikit-learn machine-learning layer that is made available via RESTful APIs that are secured with JWT based authentication.*

KEYWORDS: *Skill gap analysis, Career recommendation system, TF-IDF, Cosine similarity, Random Forest Classifier, Resume parsing, NLP, Machine learning, Personalized learning roadmap, Progress tracking.*

1. INTRODUCTION

1.1 Background

Over the past decade, the job market has transformed significantly, as a result of the impact from artificial intelligence, automation, cloud computing and fast-paced sector-wide digitalisation. The jobs which once required a tightly defined range of skills are changing within a couple of years, and new types of occupations are being created at a much faster rate than old ones are being incorporated into traditional curricula and career counselling systems. This leaves students and young professionals with two serious challenges: They have a required knowledge set that is getting larger, but they are not being taught what is being sought after in the workforce, and the competencies they learn are constantly changing. It is now widely acknowledged that this is not a personal problem, but rather a structural problem in today's labour market and that this directly affects employability, workforce readiness and long-term career satisfaction.

Traditional methods of career guidance are through aptitude tests, individual counselling and job search websites that are keyword targeted. First, no one can remember the myriad of possible personality types or interests; second, no human being can remember the skills requirements for thousands of different job types; third, a "keyword matched" job posting provides no way to measure how far the candidate's actual skill set is from the skills needed by that job; finally, no one can remember how to make that distance into a learning plan. The upshot is that job-seekers are left to their own devices to figure out which job they should apply for, or which they should overlook because a limited range of newly acquired skills would have made them available. A viable solution to overcome these constraints is natural language processing (NLP) and machine learning (ML). The resumes and job descriptions are, after all, unstructured text, and there are many techniques that can convert this text into structured skill information that can directly be compared to the skills of a target role statistically, which can lead to a prediction of what a suitable career direction would be for a candidate's competency profile; the latter is a step further, and makes career advice a continuous, data-informed process, rather than a one-time operation. To match a user to the appropriate opportunities and the appropriate learning resources, therefore, is a "composite problem" that has been tackled in the literature in different mixes of parsing, similarity matching, clustering, classification and — increasingly — generative and retrieval-augmented models.

1.2 Objectives

Motivated directly by gaps identified from literature surveyed, this paper introduces the AI-Powered Skill Gap Analysis and Career Recommendation System in which the language gap between the explicit quantification of skill-gap measurement and the narrow integration between skill-gap analysis and learning-path recommendation are bridged, and reliance on proprietary generative-AI services is avoided and replaced with transparent and reproducible parsing and matching. The proposed system is an end-to-end pipeline that will be able to extract skills from a user

provided resume or manual entry, predict the appropriate career role to be targeted by a supervised classifier, quantify the actual skill gap between the user's profile and the ideal profile for the targeted role, generate a prioritized learning roadmap from the skill gaps, and provide a personalized learning roadmap within a secure and extensible web architecture.

The rest of this paper is organized as follows. Section 2 reviews the literature across five functional themes — resume and skill data acquisition, skill representation and feature extraction, career recommendation and role prediction, skill-gap identification and roadmap generation, and explainability, personalization and progress tracking — followed by a comparative analysis of the reviewed works (Section 2.6) and the resulting research gaps (Section 2.7). Section 3 presents the proposed system's methodology: the system development framework (Section 3.1), its functional modules (Section 3.2), and its implementation details (Section 3.3). Section 4 discusses the system's applications and expected benefits. Section 5 discusses the limitations, future scope and conclusion.

2. LITERATURE REVIEW

In this section, 16 research works are reviewed, which relate to the skill-gap analysis and career recommendation using AI. The papers are not organized in alphabetical order nor in the order of when they were submitted, but rather, are organized below by the 5 functional themes that are proposed for the pipeline: skill data acquisition; skill representation and feature extraction; career recommendation and role prediction; skill-gap identification and roadmap generation; explainability, personalization and progress tracking. A paper that makes multiple contributions is discussed as to the main contribution. For ease of reference, all sixteen works are summarized in the one table (I) which is presented in Section 2.6.

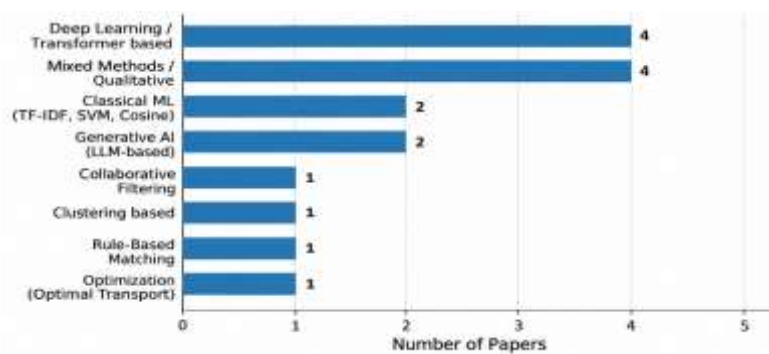


Figure 1: Distribution of core methodological approaches across the reviewed literature. Deep-learning/transformer methods and mixed/qualitative studies are the two largest clusters, followed by classical ML and generative-AI approaches.

2.1 Resume and Skill Data Acquisition

The first step in any skill-based system is to convert an unstructured resume, or free-form entry, into a structured, comparable skills list. Three studies are reviewed which address this stage directly.

Instead of a hand-coded extraction tool, Gupta and Singh [6] demonstrate a generative language model (Google Gemini) directly prompted on raw PDF textual data generated with PyPDF2 that can return a percentage match, a list of missing keywords and suggestions for improvement of a job description; a second prompt given to the generative language model extracts skills and keywords that can be used to drive live LinkedIn scraping for job suggestions. The convenience of this approach is however coupled to a paid third party that is both rate-limited and has no guarantee of a stable output schema.

Decorte et al. [11] state that skill-extraction models based on large-scale general-purpose language models are accurate but have high production costs due to their large size, and introduce a lightweight token-level contrastive bi-encoder trained with a symmetric Info NCE loss and the Skill-XL benchmark of exhaustive sentence-level ESCO skill annotations, as well as Job BERT V2 for job-title normalization. The method is still able to reduce the gap at a very small fraction of the cost of inference with large-LLM extractors, but it requires a special training corpus and compute budget beyond what might be available for a student-built system.

The Job Connect system [12] uses Tesseract OCR to recognize the scanned or image-based resume as text, and then a Skill-Based Rule Matching (SRM) algorithm tokenizes the text and scores it against a skill-keyword list; the text was not classified in the process of extracting it, and the performance of this system was reported as 85% precision at 78% recall. This pipeline suffers from degradation of noisy or handwritten scans, the authors themselves saying so.

All three papers present two different philosophies for learning skills: (a) outsource the extraction to a commercial generative model [6] and (b) build a lightweight, purpose-trained encoder [11] and a third, simpler approach of OCR

plus deterministic rule matching [12]. All three leave the possibility of depending on paid external APIs or the need for training data that exceeds the scope of student projects, or fragility from noisy scans, which led to the double manual-entry/PDF-upload design chosen for Section 3.2.1.

2.2 Skill Representation and Feature Extraction

After the raw skill terms are identified, they need to be converted into a number that is comparable. Literature surveyed shows that there are two general types of representation: sparse and interpretable vector models (SIVMs) and dense, learned embeddings.

In their paper, Chandak et al. [8] introduce a completely open-source system for parsing resumes using spaCy rule-based named-entity recognition, term frequency-inverse document frequency (TF-IDF) vectorization and Cosine Similarity comparison—the same approach the proposed system uses for matching. However, the authors state that in their actual deployment it is the similarity that is used only in an internal ranking of the job postings and not displayed to the user as a percentage.

A similar contrastive bi-encoder model (as introduced in Section 2.1 for extraction) is relevant here as a representation technique; their dense sentence representations capture contextual meaning that can not be captured by a bag-of-words model, while requiring the additional training and compute costs mentioned above.

As illustrated in this survey, the spectrum of representation techniques goes in a predictable direction: sparse TF-IDF vectors [8] offer transparency, are cheap to compute, and do not need training data, while dense embeddings [11, 13] capture semantic nuance at a real computational and infrastructural cost. There are valid reasons for the proposed system to be based on the TF-IDF representation validated in [8] and embeddings as a natural extension are discussed in Section 5.

2.3 Career Recommendation and Role Prediction Techniques

A distinct group of papers treats role recommendation either as similarity-based ranking or as supervised classification, predicting one or more suitable roles from a candidate's profile.

Prince B et al. [2] propose a "peer-mapping" collaborative-filtering scheme in which every pair of users is scored for similarity using a Naive Bayes formulation over binary skill-possession vectors, and the two most similar users are grouped so that each benefits from the other's search and application history. This surfaces only a ranked list of peers, with no individual, quantified diagnostic for the user themselves.

Wang [5] reports an Improved K-Means (IK-means) clustering method — which selects initial cluster centres to maximise inter-centre distance and filters low-density noise points — combined with the graph-based SimRank measure to compute student-to-student similarity for graduate employment recommendation, validated against UCF, SVD, LightGCN and other baselines on MAE, RMSE and clustering-quality metrics. The output, however, remains a ranked list of matched roles rather than a percentage-based gap measurement.

Mashayekhi et al. [10] introduce "congestion" — a small subset of vacancies being repeatedly surfaced to many seekers while others receive little exposure — as a marketplace-level fairness concern, and solve it with ReCon, an in-processing method that augments a base recommender's objective with an entropic optimal-transport term solved via the Sinkhorn algorithm. This optimises for exposure fairness across the marketplace rather than for individual diagnostic feedback to a single user.

Rajput et al. [4] embed a resume analyzer and job recommender, Career Craft AI, within an existing e-learning platform, training a Support Vector Machine to classify user skill profiles for grouping and comparison while content-based Cosine Similarity filtering separately drives skill-gap-aware course and job recommendations. The classifier, however, only functions meaningfully inside that parent platform's course and quiz ecosystem.

Siswipraptini et al. [7] combine an Educational Data Mining–Grounded Theory (EDM-GT) framework — Average Linkage Hierarchical Clustering over scraped job postings, plus grounded-theory-coded mappings from Myers–Briggs personality types and university subjects — with a personalized Naive Bayes classifier that predicts the most suitable job and supporting subjects, reporting 85% classification accuracy on an Indonesian computer-science student population. The mappings, however, were built and validated through subjective expert judgment in focus groups, and the classifier depends on a personality-type input many systems, including the one proposed here, do not collect.

Anghel et al. [14] decouple CV entity extraction from candidate-job classification into two independently scalable microservices: a fine-tuned BERT encoder outputs a JSON object over six ontology-matched semantic feature classes, which an XGBoost classifier with Platt scaling then scores, with SHAP explanations exposed so recruiters can manually re-weight individual feature classes. The authors report a substantial accuracy drop when the model — trained on synthetic CV-job pairs — is evaluated on real-world data, with ROC-AUC falling from 0.9 to 0.58.

Across this group, similarity-ranking approaches [2], [5], [10] stop short of a quantified, user-facing diagnostic, while classification approaches [4], [7], [14] each tie prediction to an external platform, a subjective mapping step, or a synthetic-to-real accuracy gap — observations that directly motivate the dual similarity-plus-classifier design of Sections 3.2.3 and 3.2.4.

2.4 Skill Gap Identification and Learning Roadmap Generation

A smaller number of publications move beyond recommending a role and a range of postings to seek to quantify the skills of a user and a target profile and to develop a learning plan that addresses the difference between the two.

In CareerGaide [1] Ashrafi et al. combine a deep learning based salary/income predictor, a weighted multi-field similarity function and the TF-IDF/K-means clustering of Amazon books into a single privacy preserving pipeline, producing predictions of salary and promotion, a step to detect skills gaps, and a book-based learning-path recommendation that is ordered by a helpfulness-weighted review score. The roadmap that results, however, is targeted to a single content catalogue from the outside and is not generalized to courses, certifications or project-based learning.

Braun's doctoral thesis [16] presents a prototype of skill-management that is light-weighted in terms of inputs: it infers the expected skill profile of a person based on the job role and the employer, using cross-references with LinkedIn, the European ESCO framework and the O*NET, rather than using a skills framework that is fully filled out manually. The resulting self-assessed gap is matched to available training content, as assessed through a survey of 137 participants and in two country cases. However, this gap is not based on a real-time skill vector computed against the skill, but rather on the user's reported gap, and the matching process is validated at survey level, not as a real-time, automatic feature.

Hurlburt [3] doesn't even suggest an algorithm, but instead calls for a change in the present credentialing system, more transferable-credit programs, and industry-education partnerships. This is not a functional comparison to [1] and [16] but rather serves as motivation when creating explicit, individual level tooling.

The most similar published system to the skill-gap-and-roadmap stage used in the proposed system is either [1] which constrains the learning-path content to one external catalogue, or [16]'s self-assessed, survey-driven roadmap, which is not as similar as the automatically generated, ranked set-difference against a role's required skill vector that Sections 3.2.5 and 3.2.6 target.

2.5 Explainable AI, Personalization and Progress Tracking

The final theme is work that goes beyond even a single recommendation event – work on explainability, user feedback-based adaptive learning, or the organizational context of tackling skill gaps at scale.

To garner insights that are relevant to execution in practice, Braun et al. [9] conduct a large-scale qualitative interview and thematic-analysis study of how organizations practice managing skill gaps, and they conclude that managerial and motivational factors are crucial for the success of a skill-development initiative, not just algorithmic matching. The study is non-computational and qualitative, and the software is not available in such a way that people can follow their own progress.

Nandi et al. [15] created Sankalp, a multilingual multi-agent conversational career-guidance system through reinforcement-learning fusion layer that dynamically learns its recommendation score based on whether a user accepted the same recommendation before or not, with a 90.5% and 4.71/5.0 Top-3 Hit Rate and user satisfaction respectively. This is a much more complex multi-agent architecture than the scope of the proposed system, and their adaptation is based on the acceptance of a recommendation, not based on a longitudinal skill-match percentage as is done in adaptation.

The explainability dimension raised by Anghel et al.'s explainability scores using SHAP [14] and the scores generated by Bhavesh Kumar P and Chellamani's RAG explanation of matches [13] (both of which were previously discussed) also apply here; both provide a rationale to back up a recommendation, but neither saves that rationale, or a similar numeric score, from one visit to the next for a given user.

[9] marks off the theme at one end, with a purely qualitative study of the barriers to organizational adoption, and [15] at the other end, with a fully agentic, adaptive conversational system that inspires the specific, lighter-weight form of progress tracking (persisted percent-match history per user) that Section 3.2.7 describes for the proposed system.

2.6 Comparative Analysis of Reviewed Literature

The summarized information from the reviewed literature is summarized in Table 1, organized by author/year, core technique, dataset/experimental basis, principal contribution and reported limitation. There are a number of trends that can be seen from the comparison.

Table 1: Summary of Reviewed Literature

Ref.	Author(s) / Year	Technique	Dataset / Basis	Contribution	Limitation
[1]	Ashrafi et al., 2023	CNN salary prediction + TF-IDF/K-means clustering	Resume corpus + Amazon book reviews	CareerGaide: unified salary, gap and book-based roadmap pipeline	Roadmap limited to a single external content catalogue
[2]	Prince B et al., 2023	Naive Bayes peer-mapping collaborative filtering	User search/application-history dataset	Peer-similarity-based job and course recommendation	Ranked peer list only; no quantified individual diagnostic
[3]	Hurlburt, 2024	Policy / literature synthesis (non-computational)	Secondary literature and policy sources	Institutional framing of the skills-gap problem	No individual-level computational tool

Ref.	Author(s) / Year	Technique	Dataset / Basis	Contribution	Limitation
[4]	Rajput et al., 2024	SVM classification + Cosine Similarity content filtering	E-learning platform resume + quiz data	Career Craft AI: resume analyzer and job recommender	Tied to the parent e-learning platform's ecosystem
[5]	Wang, 2024	Improved K-Means (IK-means) + SimRank graph similarity	University graduate-employment dataset	Student-to-student similarity for employment recommendation	Ranked-list output; no gap percentage
[6]	Gupta & Singh, 2024	Generative AI (Gemini) + PyPDF2 + LinkedIn scraping	Resume PDFs and live job postings	Prompt-based match %, missing-keyword and suggestion generation	Dependent on a paid, rate-limited external API
[7]	Siswipraptini et al., 2024	EDM-Grounded Theory + personalized Naive Bayes (MBTI)	Indonesian CS student dataset	Personalized job/subject prediction, 85% accuracy	Subjective expert-coded mappings; needs personality input
[8]	Chandak et al., 2024	spaCy NER + TF-IDF + Cosine Similarity	Open resume/job-posting corpus	Open-source resume parser and job recommender	Similarity used for ranking only, not shown as a percentage
[9]	Braun et al., 2025	Qualitative interviews + thematic analysis	Multi-organization interview study	“Skill Bridge” organizational skill-gap-management framework	Non-computational; no individual tracking artifact
[10]	Mashayekhi et al., 2024	Entropic Optimal Transport (ReCon / Sinkhorn)	Job-marketplace exposure logs	Congestion-aware, fairness-optimised recommendation	Optimises marketplace fairness, not individual diagnostics
[11]	Decorte et al., 2025	Token-level contrastive bi-encoder (ConTeXT-match)	Skill-XL / ESCO-annotated benchmark	Lightweight, low-cost skill-extraction encoder	Requires a dedicated training corpus and compute budget
[12]	Kumar et al., 2025	Tesseract OCR + Skill-Based Rule Matching	Scanned/image-based resumes	Job Connect: OCR-driven skill extraction, 85% precision	Degrades on noisy or handwritten scans
[13]	Bhavesh Kumar P & Chellamani, 2025	Sentence-Transformers + quantized LLM (RAG)	Resume/job-description corpus	Natural-language match explanations via retrieval-augmented generation	Explanation is qualitative, not a reproducible score
[14]	Anghel et al., 2025	Fine-tuned BERT + XGBoost + SHAP	Synthetic CV-job pair dataset	Explainable, recruiter-adjustable matching microservices	ROC-AUC drops from 0.9 (synthetic) to 0.58 (real data)
[15]	Nandi et al., 2025	SBERT + VADER + RL + knowledge graph	Multilingual conversational-agent logs	Sankalp: adaptive multi-agent career-guidance system, 90.5% Top-3 hit rate	Heavy multi-agent architecture; adapts on acceptance, not measured progress
[16]	Braun, 2026 (thesis)	Light-input skill inference + training-content matching	137-participant survey; two-nation case study	Two-input (role, employer) skill-gap inference against ESCO/O*NET	Self-reported gap; survey-scale validation, not a live feature

From Table 1 the following broad observations can be made: First, the resume- and skill-acquisition group [6], [11], [12] respectively extract the HR's resume adequately, but require a good amount of money from an external API service, a dedicated training corpus, or scan quality — none of them are free, lightweight and robust at the same time. Second, the representation and matching group [2], [5], [8], [10], [13] is good at ranking candidates, or roles, but doesn't consistently provide a quantified, user-facing indication of how similar or dissimilar the roles are, as the similarity score is not a deliverable. Thirdly, the classification based role-prediction group [4, 7, 14] all associate prediction with a third party platform, subjective expert coded mapping or dropping accuracy from mark to real. Fourth, the gap-and-roadmap group (e.g., [1], [16]) recognizes the importance of explicit, individual-level measurement but don't provide a fully formed automatically computed, role-specific roadmap, and the progress-oriented work (e.g., [9], [15]) demonstrates the need for continuity of tracking, but doesn't explicitly provide a lightweight tracking mechanism.

Table 2: Distribution of the Reviewed Literature Across Functional Themes

Functional Theme	Papers	Share of Reviewed Literature
A. Resume & Skill Data Acquisition	[6], [11], [12]	19%
B. Skill Representation & Matching	[2], [5], [8], [10], [13]	31%
C. Career Recommendation & Role Prediction	[4], [7], [14]	19%
D. Skill-Gap & Roadmap Generation	[1], [3], [16]	19%
E. Explainability, Personalization & Progress	[9], [15]	13%

Overall, the comparative analysis shows that substantial progress has been made on individual components (extraction, representation, matching and classification), but not many studies have been done that incorporate all these components into a complete system that also quantifies the gap, ranks a remediation path and continues progress over time. This observation directly inspires the RQs of Section 2.7 and the architecture proposed in Section 3.

2.7 Research Gap and Motivation

Based on the results of Sections 2.1–2.6, some gaps in the research are identified which directly influence the design of the proposed AI-Powered Skill Gap Analysis and Career Recommendation System.

Gap 1 – There is not a lot of evidence that similarity scores are discussed as a quantitative diagnostic. Matching approaches [2], [5], [8], [10] and [13] include an internal similarity/relevance score, but only report it as a ranking signal

or as a qualitative explanation, but not as a straightforward skill-match percentage that the candidate can feed into their workflow.

Gap 2 – External platforms and/or subjective inputs are linked to role prediction. These classification-based systems, such as [4] and [7] and [14], are limited to operation within a parent e-learning platform, rely on self-assessment of personality types and do not achieve high accuracy when applied to real data.

Gap 3 – Measurement of skill-gaps is either not present or is self-reported. Most of the reviewed systems require a role but do not generate a set difference against the required skills of that role, and if there is a gap in a list as estimated by some systems [16] that gap is self-assessed by the user, not extracted from a skill vector.

Gap 4 – Learning roadmaps are associated with only one content catalogue. If a roadmap is created (and if it is, it does not come from another person's list of competencies, as would be the case if one is being provided as a ranked list of missing competencies pointing towards any learning resource and not from one external, commonly accessible source, such as a book catalogue)

Gap 5 – reliance on paid or 'belt and brace' external services. Various systems [6] rely on commercial generative-AI APIs for the extraction of key information or simply matching, which affects the cost, rate limits, and external failure point that should be avoided by a self-contained academic system.

Gap 6 – No long term, longitudinal snapshot of a user's journey. Qualitative studies on organizations [9] show that the practice of continuous monitoring is required in the real world; there are adaptive systems [15] available, but none of the systems reviewed provides a light-weight, directly interpretable percent-match history a returning user can watch as it improves.

Gap 7 – Absence of one end to end integrated pipeline. Each of these steps—extraction [6],[11],[12], matching [2],[5],[8],[10],[13], classification [4],[7],[14] and gap quantification [1],[16] and progress tracking [9],[15]—has been demonstrated separately; few studies combine all these steps into one system that can be repeated.

The 7 gaps are directly mitigated by the 8 functional modules presented in Section 3 of the proposed system, and are explicitly mapped to them in Table 3.

3. METHODOLOGY

3.1 System Development Framework

Based on the research gaps discussed in Section 2.7, a web-based AI-Powered Skill Gap Analysis and Career Recommendation System is designed, which brings together the processes of processing the resume, extracting skills, TF-IDF/Cosine-Similarity matching, Random Forest role prediction, calculating and quantifying a skill-gap, generating a personalized roadmap, and persistent progress tracking in a coordinated way. The overall architecture of the system is shown in Figure 2, with each stage explained in Section 3.1 and the module descriptions of Section 3.2.

The system involves six stages of information processing, which are coordinated and shown as an overall system in Figure 2. A user provides skill information either by uploading a resume in PDF format or by completing a structured web form. This input is then sent to the Skill Extraction Module to normalize it into a structured set of skills, and sent in parallel to the Skill Matching Module (TF-IDF vectorization and Cosine Similarity against a vector of skills specific to each role, learned from the training set) and the Career Prediction Module (Random Forest Classifier trained on labelled skill profiles). A role-fusion step merges the similarity ranking and the prediction from the classifier to a common target role that is cross checked. The Skill Gap Analysis Engine then calculates the set difference between the target role's ideal skill vector and the user's extracted skill vector, and then the Personalized Learning Recommendation Module types and prioritizes missing skills in a roadmap. Every analysis — the match percentage, predicted role, missing-skill list and roadmap — is persisted by the User Progress Tracking Module to allow a returning user to see that their match percentage is improving as time goes on.

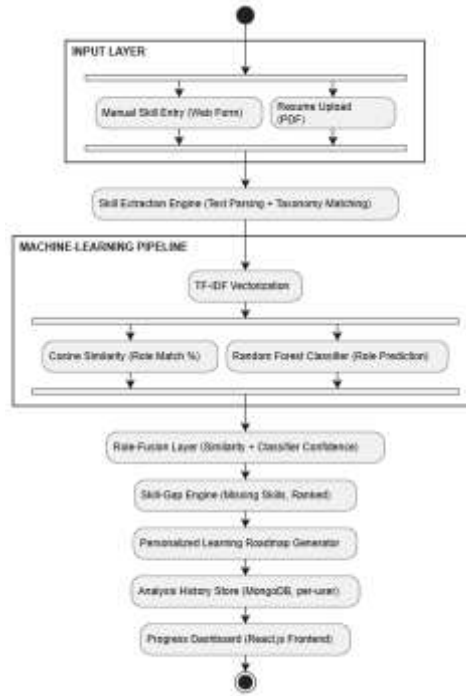


Figure 2: Complete system architecture of the AI-Powered Skill Gap Analysis and Career Recommendation System, from input acquisition through matching, classification, gap quantification, roadmap generation and progress tracking.

3.2 System Modules and Functional Components

3.2.1 Resume Processing Module

Unlike the OCR-dependence of [12] and the API-dependence of [6], Resume Processing Module supports two inputs, one structured web form and the other a resume upload in PDF, both with equal weightage, to avoid the limitations of having only one path to enter resumes — the parsing path. The module receives a PDF and extracts raw text using a PDF text-extraction library, removes non-textual layout artifacts and passes the resulting plain text to the Skill Extraction Module. This step is intentionally kept simple and without dependencies and is based on digitally generated PDF text, since resumes currently don't use OCR on scanned images, as discussed in Section 5.

3.2.2 Skill Extraction Module

The Skill Extraction Module takes raw resume text or manually entered skills and transforms it into a structured, comparable set of skills by processing it through a deterministic natural-language-processing pipeline: lowercasing and punctuation removal, tokenization, stop-word removal, lemmatisation/normalisation, and finally matching keywords with a maintained skill taxonomy with support for synonyms such as "JS", "Javascript", and mapping the two to the same canonical skill token. It is a much more flexible design than the pure keyword-list matching of [12] as it can be applied to a wide range of resume formats, as normalization and lemmatization are performed before the matching. It does not need an external API as in [6] and does not require a dedicated training corpus like in [11].

3.2.3 Skill Matching Module

The Skill Matching Module encodes the user's extracted skills and the skill sets that are available for each role in a common corpus as TF-IDF vectors, and calculates the Cosine Similarity between both vectors. The value computed by an equivalent TF-IDF/Cosine pipeline, which is a number between 0 and 1, independent of the length of the vector, is surfaced to the user as a skill-match percentage for each candidate role, along with being used to rank candidates, unlike [8] which only uses it to rank job postings internally. This is the specific mechanism of changing a similarity score into an individually actionable diagnostic, not just a ranking signal, or a narrative explanation, which is the Gap 1 identified in Section 2.7.

3.2.4 Career Prediction Module

The Career Prediction Module works in parallel with the Skill Matching Module and trains a Random Forest Classifier which consists of a set of decision trees trained on a bootstrapped subset of a labelled skill-profile dataset, where the decision trees are combined by the majority voting principle (Figure 3). The model used to remove the platform lock-in of [4] and the subjective personality-mapping step in [7], while also avoiding the synthetic to real domain transfer gap reported in [14] by training and testing the model with the same feature representation as the time of inference. The two independent techniques cross check each other: the classifier's per-class probabilities are fused in a role-fusion step with the Cosine Similarity ranking from Section 3.2.3, a step that does not stand in isolation, but rather directly addresses Gap 2.

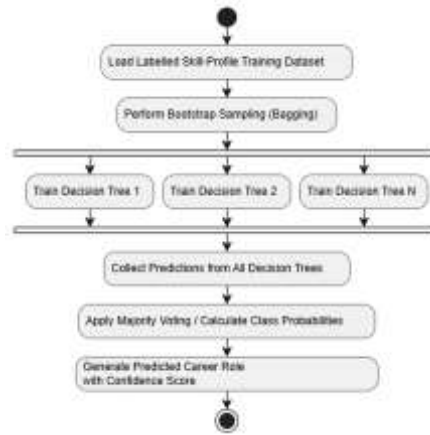


Figure 3: Random Forest career-prediction architecture.

3.2.5 Skill Gap Analysis Engine

After the role-fusion step is finished, the Skill Gap Analysis Engine performs explicit set-difference between the ideal skill vector of the target role and the skill vector extracted from the user's profile to get the list of skill elements that the user is missing, and ranks this list according to each missing skill's TF-IDF weight in the target role's profile (Figure 4). This is automatically computed per-role gap, as opposed to [1] which is based on a roadmap of a single catalogue or [16] which is based on a self-assessed survey based gap, and directly answers Gap 4 from Section 2.7.

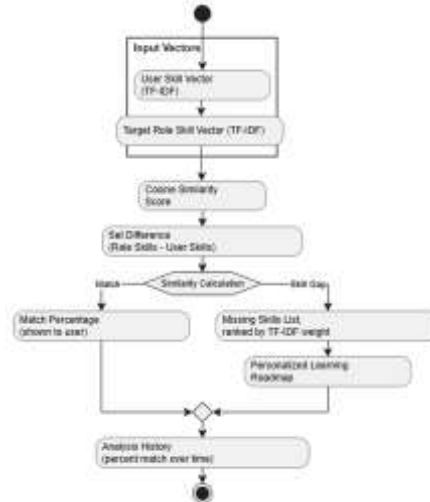


Figure 4: Skill-gap identification and missing-skill analysis flow.

3.2.6 Personalized Learning Recommendation Module

The ranked missing-skills list generated in Section 3.2.5 is not directed to a single external catalogue of content as in [1] or to a survey validated, self-reported match as in [16], but rather bundled into a structured, prioritized learning roadmap, by the PLRM. The roadmap is created directly from the computed skill gap and in principle can refer to any type of learning resource, such as course, certification, or project which covers Gap 4.

3.2.7 User Progress Tracking Module

Each analysis a user performs - such as calculating the match percentage, identifying missing skills, and constructing a roadmap - is stored in a per-user collection of analysis-history in MongoDB. This history is read by the User Progress Tracking Module, which provides a dashboard, displaying progress of a user's match percentage over a succession of analyses, providing the level of continuity that the organizational study in [9] characterizes as a needed feature.

Lightweight adaptive agent architecture, as in [15], is not necessary for real-world adoption, given a lightweight directly interpretable metric. This directly addresses Gap 6.

Table 3: Mapping of Research Gaps to Proposed System Components

Gap	Addressed By Module	Section
G1: Similarity score not surfaced as a diagnostic	Skill Matching Module (TF-IDF + Cosine Similarity shown as match %)	3.2.3

Gap	Addressed By Module	Section
G2: Role prediction coupled to platform or subjective inputs	Career Prediction Module (independent Random Forest Classifier)	3.2.4
G3: Skill-gap measurement absent or self-reported	Skill Gap Analysis Engine (computed set-difference)	3.2.5
G4: Roadmap tied to a single content catalogue	Personalized Learning Recommendation Module (ranked, role-specific roadmap)	3.2.6
G5: Dependence on paid external services	Resume Processing + Skill Extraction Modules (self-contained NLP pipeline)	3.2.1, 3.2.2
G6: No persisted, longitudinal progress view	User Progress Tracking Module (per-user analysis history)	3.2.7
G7: Lack of a single end-to-end pipeline	Complete eight-module system architecture	3.1

Table 3: shows that each of the seven research gaps identified in Section 2.7 is addressed by a specific, named module of the proposed eight-module architecture, so no gap is left unmapped to a concrete design decision.

3.3 Implementation Details

3.3.1 System Workflow

The entire data flow of the developed system is depicted in Figure 5. The React.js front end communicates with an underlying REST API using Node.js/Express.js, which in turn interface with a Python/Scikit-learn microservice for skill-vector computation, similarity matching and role classification, and all persistent data—user accounts, extracted skill profiles, role definitions, analysis history—is stored in MongoDB. The four core MongoDB collections are outlined in Figure 6 and the three tiers are deployed.

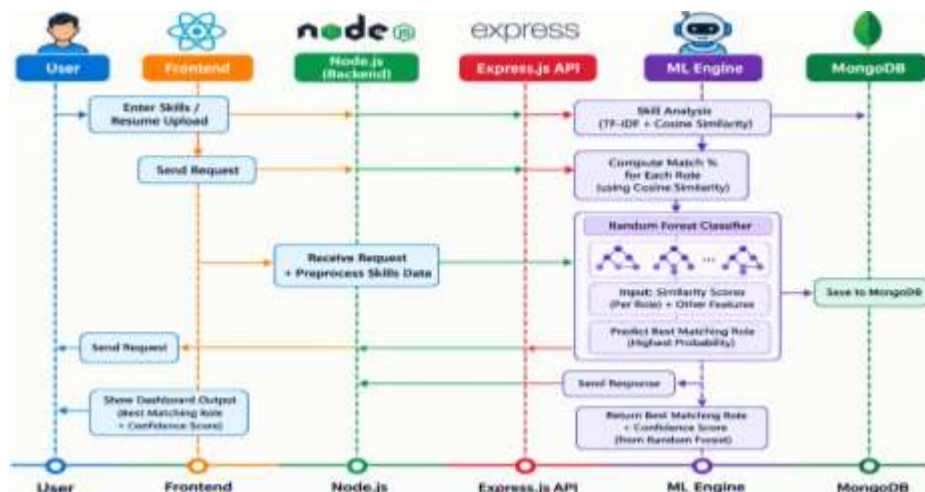


Figure 5: Complete data-flow diagram of the implemented system.

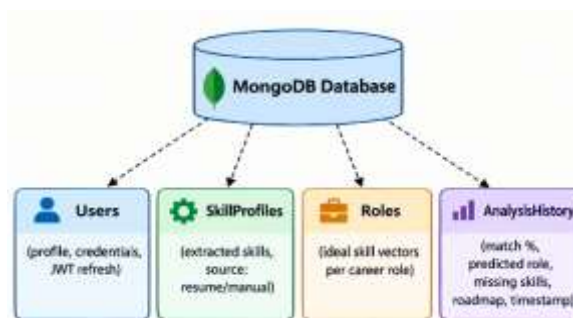


Figure 6: Database architecture showing the four core MongoDB collections.

3.3.2 Software and Hardware Requirements

The development environment is comprised of Node.js (LTS release) and npm for the front end and back end, Python 3.10+ with pip for the machine-learning microservice, and an instance of MongoDB (local or Atlas) for persistence. The proposed pipeline relies on classical machine-learning techniques (TF-IDF, Cosine Similarity, Random Forest), unlike the Transformer- or LLM-based approaches reviewed in Section 2, avoiding the need of a GPU during development or inference, which helps in keeping the cost low for development and deployment.

3.3.3 Dataset Description

The system is based on two datasets. A skill taxonomy is a carefully curated list of core, domain-specific (programming language, framework, data-science tools, soft skills) and synonym-aware list of skill terms. The Skill Extraction Module uses this taxonomy to match keywords. Second, a labelled skill-profile database, where every row would be a set of skills - corresponding to a career role, is used to create the ideal TF-IDF skill vector for each career role in Skill Matching Module and to train the Random Forest Classifier in Career Prediction Module. Both datasets are stored as structured, versioned files and new roles and skills can be added without having to train the entire pipeline from scratch.

3.3.4 Machine Learning Methodology

The user and the role vectors share a common feature space prior to calculating the Cosine Similarity, as the TF-IDF vectorizer is fit on the combined corpus of all skill-profiles (including the incoming user skill-set). The labelled skill-profile data is used to train the Random Forest Classifier, while the held-out test split is used to estimate classification accuracy, precision, recall and F1-score per role, both the number of trees and the maximum depth of the trees is optimized through cross validation, optimizing the number of trees while avoiding overfitting in a relatively small curated set of data points. The same vectorizer fitted at training time and the same classifier trained at training time are invoked for a new user's skill set at the time of inference and the results of the vectorizer (match percentage per role) and the classifier (predicted role probability) are fused together during the role fusion step of Section 3.2.4.

3.3.5 Database Design

The MongoDB schema separates the concerns into four collections: one for Users (the authentication credentials and profile metadata), one for SkillProfiles (the structured skill set extracted for each user, including its source, resume or manual entry), one for Roles (the set of maintained career roles and their ideal skill vectors), and one for AnalysisHistory (one document per analysis run, storing the computed match percentage, predicted role, ranked missing-skill list, and generated roadmap, as well as a timestamp). This separation allows for the Roles collection to be updated as new roles are added, without interacting with historical user data.

3.3.6 API Communication and Security

The application uses React.js as a Frontend, with the Node.js/Express layer interfacing with the Python ML microservice via a private internal API for vectorization, similarity computation, and classification to provide a RESTful JSON API to the Frontend. JSON Web Tokens (JWT) are used for user authentication and session management, and passwords are not stored in plain text, but rather are encrypted before they are stored. To defend against malformed or oversized input to the extraction pipeline, all client-server traffic is expected to be done through HTTPS, and input validation is done on the server side of the manual skill-entry form and uploaded resume files.

3.3.7 Testing Methodology

The implementation is checked at three levels. A series of unit tests are performed for the individual normalization steps in the Skill Extraction Module and for the TF-IDF/Cosine Similarity computation in isolation, with small, hand-built sets of known expected similarity scores. Integration tests test the entire pipeline: the sample resume is uploaded, persisted and analyzed, and then the resulting analysis-history record is persisted, to ensure that data flows correctly from the Node.js API to the Python ML microservice. Finally, informal user and acceptance testing with sample resumes and manually entered skill sets is used to sanity check the predicted roles, computed match percentages and generated roadmaps before wider use.

Table 4: Implemented Technology Stack

Layer	Component	Role
Frontend	React.js	Responsive user interface for skill entry, resume upload and dashboard display
Backend	Node.js / Express.js	REST API, request routing, authentication middleware
Backend	JWT + bcrypt	Token-based authentication and password hashing
ML Layer	Python 3.10+	Core machine-learning service runtime
ML Layer	Scikit-learn	TF-IDF vectorization, Cosine Similarity, Random Forest Classifier
ML Layer	PDF text-extraction library	Resume text extraction from uploaded PDFs
Database	MongoDB	Users, Skill Profiles, Roles and Analysis History collections
Database	Mongoose ODM	Schema modelling and query interface for MongoDB
DevOps	REST / HTTPS	Client-server and service-to-service communication

Table 4: lists the concrete technology choice behind every layer of the architecture shown in Figures 5, from the React.js presentation layer down to the MongoDB persistence layer.

4. RESULTS

4.1 System Capabilities and Research-Gap Coverage

As shown in Table 3, there is a module in the proposed architecture of eight modules for each research gap identified in Section 2.7, meaning there is no research gap which does not have a module mapped to a specific design decision.

4.2 Applications and Expected Benefits

The system allows students and early career job seekers to have multiple, student-driven career-counselling experiences rather than one, and to know what skills are a match to one or more job they are interested in pursuing, and what skills they are lacking, so they can revisit their analysis of their skills as they acquire new skills. The aggregated (anonymised) skill-gap output per cohort of learners of the academic institution and/or training provider could identify areas of skills that are being systematically under-represented in the training programme, compared to the skills that the learners are looking for when applying for the job that they are interested in. The corporate L&D team may be able to leverage the pipeline used for the skills gap approach and roadmap for internal role definition, so that employees can be upskilled or plans for internal mobility can be made rather than an external hire. The core matching and prediction component of the system does not rely on a large neural network, but rather TF-IDF, Cosine Similarity and Random Forest classification, which makes it cheap and easily runnable, increasing the ease of use for smaller institutions and individuals with limited computing resources.

5. CONCLUSION

Limitations. This is the current implementation, and there are several things to be aware of. The Semantic Similarity between related skills with different terms is another aspect where the TF-IDF representation used in the Skill Matching Module suffers as it is a bag-of-words representation and does not capture the semantic similarity between related but differently worded skills, such as "deep learning" and "neural networks" unless the two skills are explicitly linked in the skill taxonomy, a problem also experienced by [4] and [8] and partially addressed by dense embedding techniques such as [11] and [13] at higher computational cost.

Second, the accuracy of the Skill Matching and Career Prediction Modules depends on the size and quality of the curated role skill-profile dataset (as described in Section 3.3.3): In this case, the small size, as well as the limited breadth of the dataset, limit the generalisation to less common career roles, which corresponds to the dataset-size limitations reported in [7] and [14]. Third, the Resume Processing Module is only able to process digitally-generated PDF resumes, and is not yet capable of OCR. Thus, scanned or image-based resumes, as dealt with by [12] with its own limitations, are not supported. Fourth, the list of skill vectors that are supported along with the list of careers is not derived from a real time stream of jobs from the job market, but is developed and maintained by the developers: this implies that the system will not automatically update to include new careers and/or changes to the skill vectors that are required by the job-market the way a system connected to a real-time job-portal API would. Last but not least, the present solution allows to write resumes and skill entries in English language; multilingual support, such as that discussed in [15] is for future work.

Future Scope. Further research is needed to overcome the identified limitations by adding sentence embeddings to the TF-IDF or substitute it with TF-IDF to achieve more semantic skill matching [11, 13]. Integrating with live job-portal APIs [6] and [12] could automatically update the role-skill profile, and using OCR for resume processing [12] would help with scanned resumes. Anonymised skill-gap information at the regional or institutional level could also help in trend analysis for the labour market [3]. Linking to external Learning Management Systems (LMS) may also make the personalized roadmap more trackable and enrollable learning paths. All the reviewed studies focus on one of the several steps in the proposed system, namely resume processing [6, 11, 12], skill matching [2, 5, 8, 10, 13], predicting role [4, 7, 14], skill gap analysis [1, 16], policy level analysis [3] and tracking progress [9, 15] without covering all the steps of the proposed system. The proposed AI-Powered Skill Gap Analysis and Career Recommendation System combines the existing technique of TF IDF, Cosine Similarity, and Random Forest in React.js, Node.js/Express.js, MongoDB, and Python/Scikit-learn architecture. The contribution, however, is not simply in the individual algorithms, or their combination, but in their end-to-end composition—the Cosine Similarity yields a quantified “skill-match” score, while the Random Forest learns a new skill's best position independently, and the set-difference analysis creates a personal learning roadmap that is tracked over time.

Therefore, the system offers the basis for subsequent integration of semantic matching, real-time information about the labour market, the process of a resume by OCR, a connection with LMS, and labour-market analytics, and extends functionality identified in the reviewed literature.

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